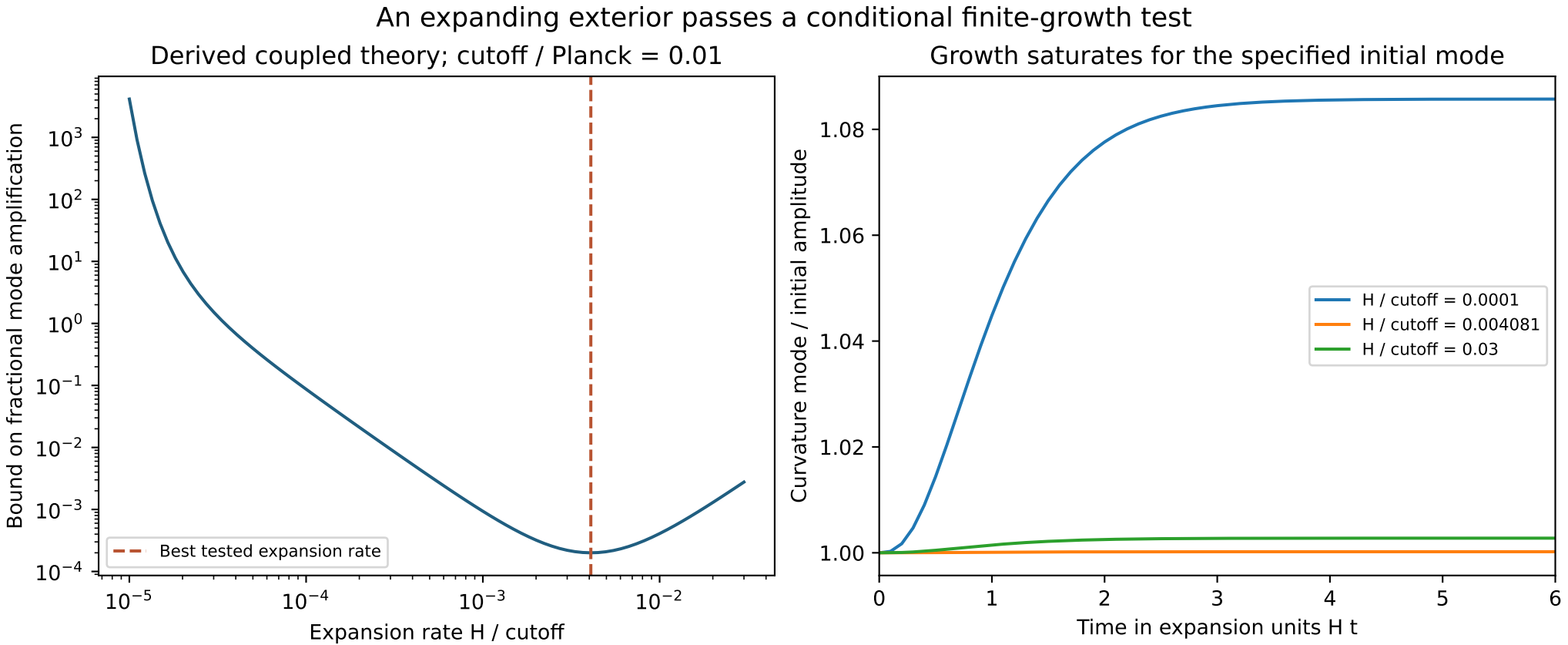
An expanding exterior passes a conditional finite-growth test
Derived coupled theory; cutoff / Planck = 0.01
Bound on fractional mode amplification
Best tested expansion rate
Expansion rate H / cutoff
Growth saturates for the specified initial mode
Curvature mode / initial amplitude
H / cutoff = 0.0001
H / cutoff = 0.004081
H / cutoff = 0.03
Time in expansion units H t
Added vacuum energy is required. No localized negative-energy source, nonlinear stability, material or time passage established.